Worksheet #5

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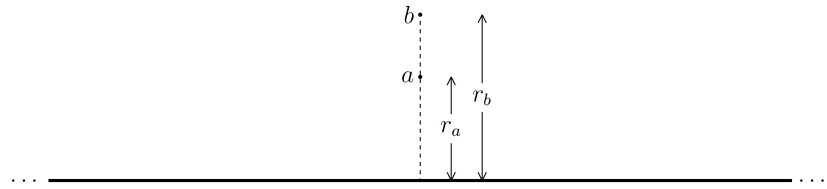
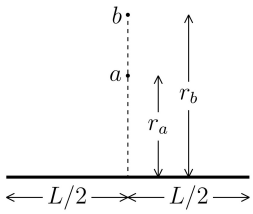
1. Answer the following:

   (a) Calculate the potential difference $V_b - V_a$ due to the infinite rod of uniform charge density , as shown below. Perform this calculation by directly integrating the electric field over a path between a and b.
   Two points a and b, distances r_a and r_b, respectively, away from a long rod



   (b) Calculate the potential difference $V_b - V_a$ due to the rod of length L and uniform charge density λ, as shown below. Perform this calculation by calculating Va and Vb directly from the charge distribution and subtracting.
   Two points a and b, distances r_a and r_b, respectively from the center of a rod of length L



   (c) Show that your answer to part b matches your answer to part a in the $\lim_{L\to\infty}$.

1.(a)

$$
\begin{aligned}
V_b - V_a &= -\int_{r_a}^{r_b} E dr \\
&= -\int_{r_a}^{r_b} \frac{\lambda}{2\pi\epsilon_0 r} dr \\
&= -\frac{\lambda}{2\pi\epsilon_0} \int_{r_a}^{r_b} \frac{1}{r} dr \\
&= -\frac{\lambda}{2\pi\epsilon_0} \ln r \Big|_{r_a}^{r_b} \\
&= -\frac{\lambda}{2\pi\epsilon_0} (\ln r_b - \ln r_a) \\
&= -\frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}
\end{aligned}
$$

1.(b)

$$
\begin{aligned}
V(r) &= \int_0^L E dr \\
&= \int_0^L k_e \frac{\lambda dx}{\sqrt{r^2 + x^2}} \\
&= k_e \lambda \int_0^L \frac{1}{\sqrt{r^2 + x^2}} dx \\
&= k_e \lambda \ln \left(\left| \sqrt{x^2 + r^2} + x \right| \right) \Big|_0^L \\
&= k_e \lambda \Big(\ln \left(\sqrt{L^2 + r^2} + L \right) - \ln \left(\sqrt{0^2 + r^2} + 0 \right) \Big) \\
&= k_e \lambda \ln \left(\frac{\sqrt{L^2 + r^2} + L}{r} \right)
\end{aligned}
$$

$$
\begin{aligned}
V_b - V_a &= V(r) \big|_a^b \\
&= k_e \lambda \ln \left(\frac{\sqrt{L^2 + r_b^2} + L}{r_b} \cdot \frac{r_a}{\sqrt{L^2 + r_a^2} + L} \right) \\
&= k_e \lambda \ln \left(\frac{\sqrt{L^2 + r_b^2} + L}{r_b} \cdot \frac{r_a}{\sqrt{L^2 + r_a^2} + L} \right) \\
&= k_e \lambda \ln \left(\frac{r_a (\sqrt{L^2 + r_b^2} + L)}{r_b (\sqrt{L^2 + r_a^2} + L)} \right)
\end{aligned}
$$

1.(c)

part b

$$
\begin{aligned}
V_b - V_a &= \lim_{L \to \infty} V(r) \big|_a^b \\
&= \lim_{L \to \infty} k_e \lambda \ln \left(\frac{\sqrt{L^2 + r^2} + L}{r} \right) \Bigg|_a^b \\
&= k_e \lambda \ln \left(\frac{2L}{r} \right) \Bigg|_a^b \\
&= 2 k_e \lambda \ln \left(\frac{L}{r} \right) \Bigg|_a^b \\
&= 2 k_e \lambda \ln \left(\frac{L}{r_b} \cdot \frac{r_a}{L} \right) \\
&= 2 k_e \lambda \ln \frac{r_a}{r_b} \\
&= \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_a}{r_b}
\end{aligned}
$$

part a

$$
\begin{aligned}
V_b - V_a &= -\frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a} \\
&= \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_a}{r_b}
\end{aligned}
$$

The answers for these two parts appears to agree